



Curve Sketching

Section

4.4

A. Guidelines for sketching a curve

1. Domain
2. Intercepts (x -intercepts and y -intercepts)
3. Symmetry (Odd, even, or periodic functions)
4. Asymptotes
5. Intervals of Increase and Decrease
6. Maximum and Minimum Values
7. Intervals of Concavity

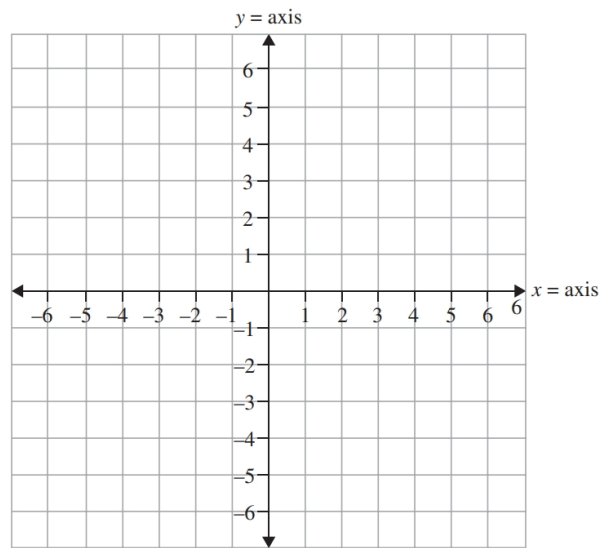
Examples

1. (video) $f(x) = \frac{x^2}{x^2 - 4}$. Sketch the curve using the guidelines 1 to 7.

1. Domain

2. Intercepts (x -intercepts and y -intercepts)

3. Symmetry (Odd, even, or periodic functions)



4. Asymptotes

5. Intervals of Increase and Decrease

6. Maximum and Minimum Values

7. Intervals of Concavity

B. Guidelines for sketching a function given a sketch of its derivative

1. Find all intervals where the function is increasing and decreasing.
2. Find all intervals where the function is concave up and concave down.
3. Sketch a function that has these characteristics (there are many graphs possible).

$f(x)$ Maximum and Minimums $\leftrightarrow f'(x)$ roots

$f(x)$ Increasing $\leftrightarrow f'(x) > 0$

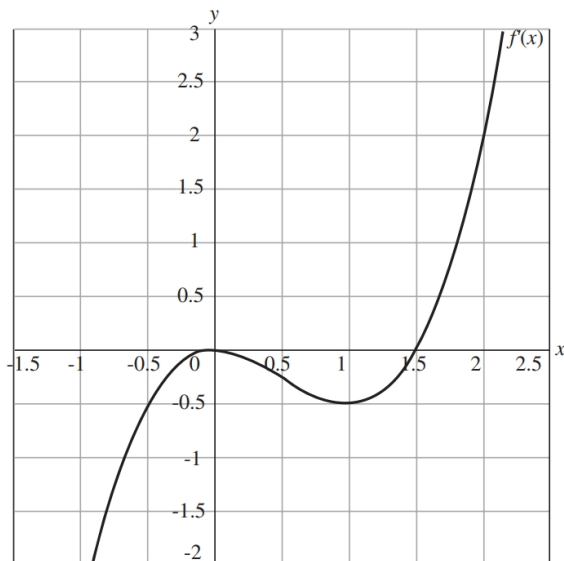
$f(x)$ Decreasing $\leftrightarrow f'(x) < 0$

$f(x)$ Inflection points $\leftrightarrow f'(x)$ Maximum and Minimums

$f(x)$ Concave up $\leftrightarrow f'(x)$ increasing

$f(x)$ Concave down $\leftrightarrow f'(x)$ decreasing

2. Given the graph of $f'(x)$, determine whether the following conditions are true



a. f is concave downward on the interval $(-\infty, 0)$

b. f has a local maximum at $x = 0$

c. f is decreasing on the interval $(-\infty, 1.5)$

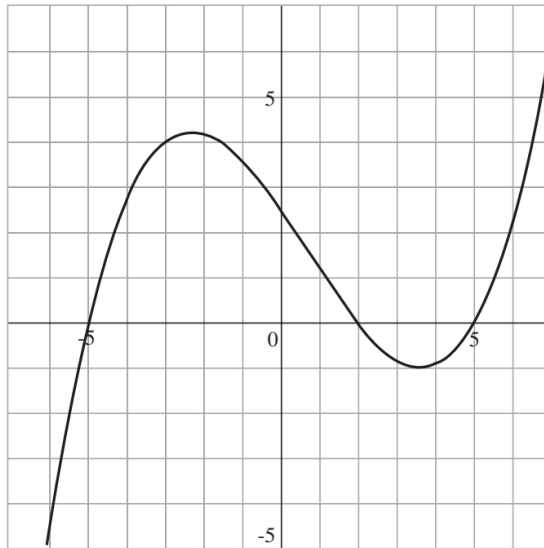
d. f has an inflection point at $x = 0$

e. f is decreasing on the interval $(0, 1)$

f. f is increasing on the interval $(-\infty, 1.5)$

3. For each of the following is the graph of the derivative $f'(x)$ of a function. State where the original function $f(x)$ is increasing, decreasing, has max and mins, concave up and down, inflection points and then sketch $f(x)$.

a.



b.

